Solution

The ellipse is symmetric about both the axes. The major axis lies along x-axis. The region to be revolved is sketched as in Fig.9.42.

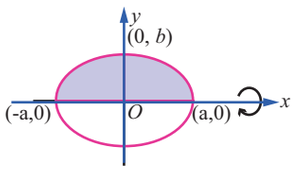


Fig. 9.42

Hence, the required volume is given by

$$V = \pi\int_{-a}^{a} y^2 dx = \pi\int_{-a}^{a} \frac{b^2}{a^2}\left(a^2 - x^2\right)dx$$

$$= \frac{2\pi b^2}{a^2}\int_0^a \left(a^2 - x^2\right)dx \text{ , since the integrand is an even function.}$$

$$= \frac{2\pi b^2}{a^2}\left(a^2x - \frac{x^3}{3}\right)_0^a = \frac{2\pi b^2}{a^2}\left(a^3 - \frac{a^3}{3}\right) = \frac{2\pi b^2}{a^2}\left(\frac{2a^3}{3}\right) = \frac{4\pi ab^2}{3}$$

Note

If the region bounded by ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is revolved about the y-axis, then the volume of the solid of revolution is $\frac{4\pi a^2 b}{3}$. The solid is called an ellipsoid.

Example 9.67

Find, by integration, the volume of the solid generated by revolving about y-axis the region bounded between the parabola $x = y^2 + 1$, the y-axis, and the lines $y = 1$ and $y = -1$.

Solution

The parabola $x = y^2 + 1$ is $y^2 = x - 1$. It is symmetrical about x-axis and has the vertex at $(1,0)$ and focus at $\left(\frac{5}{4},0\right)$. The region for revolution is shaded in Fig.9.43. Hence, the required volume is given by

$$V = \pi\int_{-1}^{1} x^2 dy$$

$$= \pi\int_{-1}^{1} \left(y^2 + 1\right)^2 dy$$

$$= 2\pi\int_0^1 \left(y^4 + 2y^2 + 1\right)dy \text{ , since the integrand is an even function.}$$

$$= 2\pi\left(\frac{y^5}{5} + 2\frac{y^3}{3} + y\right)_0^1 = 2\pi\left(\frac{1}{5} + \frac{2}{3} + 1\right) = \frac{56}{15}\pi \text{ .}$$

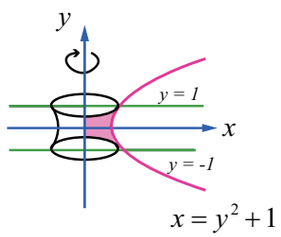


Fig. 9.43

Example 9.68

Find, by integration, the volume of the solid generated by revolving about y-axis the region bounded between the curve $y = \frac{3}{4}\sqrt{x^2 - 16}$, $x \geq 4$, the y-axis, and the lines $y = 1$ and $y = 6$.

Solution

We note that $y = \frac{3}{4}\sqrt{x^2 - 16} \Rightarrow \frac{x^2}{16} - \frac{y^2}{9} = 1$. So, the given curve is a portion of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ between the lines $y = 1$ and $y = 6$ and it lies above the x-axis.

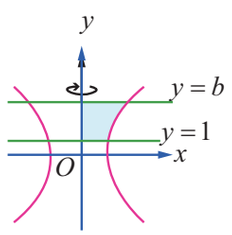


Fig. 9.44

The region to be revolved is sketched in Fig.9.44.

Since revolution is made about y-axis, we write the equation of the

Chapter 9 Applications of Integration.indd 138 17-02-2020 17:08:30